



The small inductive dimension of finite lattices through matrices

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Received: 11 April 2022 / Revised: 10 December 2022 / Accepted: 2 February 2023

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Abstract

Dimensions of partially ordered sets, lattices and frames have attracted the interest of various researches (see for example Sancho de Salas and Sancho de Salas 1991; Vinokurov 1966; Dushnik and Miller 1941; Hegde and Jain 2007; Trotter 1975; Boyadzhiev et al. 2018; Dube et al. 2015, 2017; Georgiou et al. 2016; Hai-feng et al. 2017; Brijlall and Baboolal 2008, 2010; Boyadzhiev et al. 2019). For example, the order, Krull, covering dimension, and the quasi covering dimension are some of these dimensions that have been studied extensively. Especially, finite partially ordered sets and finite lattices are the main axes of this Dimension Theory, investigating new results and characterizations for their dimensions. The matrix algebra plays an essential role in these studies, considering incidence and order matrices. Also, based on these researches, algorithms which compute these dimensions have been investigated. Since the chapter on dimensions of partially ordered sets continues to attract the interest, the small inductive dimension is a new notion that has been defined firstly for regular frames in Brijlall and Baboolal (2008). In this paper, we study the small inductive dimension for finite lattices using matrices. For that, we study the meaning of the pseudocomplement and the lattice $\uparrow x$, where x is an element of a finite lattice. Based on these investigations we present an algorithmic procedure for the matrix computation of the small inductive dimension of an arbitrary finite lattice.

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Keywords Order matrix · Incidence matrix · Algorithmic procedure · Small inductive dimension · Finite lattice

Mathematics Subject Classification 54F45 · 06A07

1 Introduction and preliminaries

The order, Krull, and covering dimension are dimensions that have been studied in the view of matrix algebra for finite posets and finite lattices (see for example Boyadzhiev et al. 2018; Dube et al. 2017; Georgiou et al. 2016). In Brijlall and Baboolal (2008, 2010) the notion of the small inductive dimension for regular frames was defined and properties of this dimension were investigated.

In this paper, based on the notion of the small inductive dimension, we study this dimension for finite lattices using matrices. Especially, using the order matrix and the incidence matrix, we present an alternative approach to this dimension. An algorithmic procedure for the computation of the small inductive dimension of an arbitrary finite lattice completes our study.

The motivation for this investigation follows by the continuous development of applications of Linear Algebra and Discrete Mathematics on pure theoretical computer science. Many topological notions, and specifically topological dimensions, have been studied through a discrete aspect. That is, combining classical topological notions with that of Linear Algebra and Discrete Mathematics we can compute mathematical results by studying algorithmic procedures.

Such a study began from the investigation of finite spaces and Alexandroff spaces as topological models in Digital Topology to specify image analysis, image synthesis, computer graphics and algorithms. More precisely, finite sets are one of the most important “keys” to succeed in alternative investigations of topological notions since they have many applications for computer science, algorithms, programming languages, cryptography and software development. This fact tends us to pay attention to the meaning of “finite”. Partially ordered sets and especially lattices are used also to investigate more properties of digital images and their topological dimensions are the main axes of this investigation.

Based on the above discussion, this paper is devoted to finite lattices, its kind of dimension, the small inductive dimension, and a special investigation of this dimension. It is formed as follows. In Sect. 2, we study the meaning of the pseudocomplement x^* of any element x of arbitrary finite lattice L using matrices. An algorithm which finds this element gives a different approach of this study. In Sect. 3, we study the finite lattice $\uparrow(x \vee x^*)$, where x is an arbitrary element of a finite lattice, presenting also an algorithm which determines the lattice $\uparrow(x \vee x^*)$ through order and incidence matrices. Finally, in Sect. 4, we study the combination of the small inductive dimension with the so-called minimal covers and present an algorithm which computes the small inductive dimension of an arbitrary finite lattice using matrices.

First of all, we recall basic definitions and notations which will be used in the following sections.

A set L endowed with a partial order $\leq$ is called a *partially ordered set* (usually, *poset*). A poset L is called *lattice* if every finite subset of it has supremum and infimum in L .

Moreover, two elements x and y of a poset $(L, \leq)$ are said to be *comparable* if $x \leq y$ or $y \leq x$. A subset of a poset $(L, \leq)$ in which every pair of elements is comparable is called a

chain. An *antichain* is a subset of a poset $(L, \leq)$ in which any two distinct elements in the subset are not comparable. If x and y are not comparable, we shall write $x \parallel y$.

We mention that in the rest of the study, we refer to a finite lattice L with n elements, $n \geq 2$. It will be denoted by $L = \{x_1, \dots, x_n\}$, where $x_1 = 0_L$ and $x_n = 1_L$, are the bottom and the top elements of L respectively.

If L is a finite lattice and $x \in L$, then we write $\uparrow x = \{y \in L : x \leq y\}$ and $x^* = \bigvee \{y \in L : y \wedge x = 0_L\}$. In the case where $x^* \wedge x = 0_L$, that is $x^* = \max\{y \in L : y \wedge x = 0_L\}$, the element x^* is called *pseudocomplement* of x . Every finite distributive lattice is pseudocomplemented. Although the lattices we deal with in this paper are not generally distributive, we will adopt the asterisk notation and the term “pseudocomplement” to have the same terminology as in Brijlall and Baboolal (2008, 2010). We state that the poset $\uparrow(x \vee x^*)$, where $x \in L$, is also a finite lattice.

A subset V of a finite lattice L is called a *cover* of L if $0_L \notin V$ and $\bigvee V = 1_L$. A cover U of a finite lattice L is called a *refinement* of a cover V of L , writing $U \succ V$, if for each $u \in U$, there exists $v \in V$ such that $u \leq v$.

Especially, a cover V of a finite lattice L is called *minimal* if $V \subseteq U$, for every refinement U of V .

The small inductive dimension was defined in Brijlall and Baboolal (2008) for regular frames and properties of this dimensions have been studied in Brijlall and Baboolal (2008, 2010). Here we study the small inductive dimension for the class of finite lattices.

Especially, if L is a finite lattice, then the *small inductive dimension*, ind , of L is defined as follows: $\text{ind}(L) \leq k$, where $k \in \{0, 1, \dots\}$, if for every cover V of L , there exists a cover U of L such that U is a refinement of V and $\text{ind}(\uparrow(u^* \vee u)) \leq k - 1$, for every $u \in U$. In particular, $\text{ind}(L) = k$, where $k \in \{0, 1, \dots\}$, if $\text{ind}(L) \leq k$ and $\text{ind}(L) \not\leq k - 1$. Moreover, we define $\text{ind}(L) = -1$ if and only if $L = \{0_L\}$.

2 Pseudocomplements in finite lattices with matrices

As we can see in the definition of the small inductive dimension, the notion of the pseudocomplement plays an essential role. Therefore, in this section we study it using matrices. Especially, we present a characterization of this notion using the so-called incidence and order matrices. Based on these results, we provide an algorithmic procedure, which finds the pseudocomplement of any element in an arbitrary finite lattice.

Let $(L, \leq)$ be a finite lattice, where $L = \{x_1, \dots, x_n\}$. Then we shall use the following two matrices (see for example (Boyadzhiev et al. 2018; Georgiou et al. 2016)):

(1) the $n \times n$ matrix $T_L^{\leq} = (t_{ij})$, where

$$t_{ij} = \begin{cases} 1, & x_i \leq x_j \\ 0, & \text{otherwise,} \end{cases}$$

which is called the *incidence matrix* of L and

(2) the $n \times n$ matrix $\mathbf{A}_L^{\leq} = (a_{ij})$, where

$$a_{ij} = \begin{cases} 1, & \text{if } i = j \\ 2, & \text{if } x_i < x_j \\ -2, & \text{if } x_j < x_i \\ 0, & \text{if } x_i \parallel x_j \end{cases}$$

which is called the *order matrix* of L .

In what follows, for every finite lattice $(L, \leq)$ we have the incidence matrix $T_L^\leq = (t_{ij})$ and the order matrix $A_L^\leq = (a_{ij})$. We denote the n columns of the incidence matrix $T_L^\leq$ by $C_1(T_L^\leq), \dots, C_n(T_L^\leq)$ and the n rows of the order matrix $A_L^\leq$ by $r_1(A_L^\leq), \dots, r_n(A_L^\leq)$. Moreover, if we observe that α is an element of a matrix A (respectively, α is not an element of a matrix A) we shall write $\alpha \in A$ (respectively, $\alpha \notin A$).

We state that the combination of the incidence and order matrices of a finite lattice $L = \{x_1, \dots, x_n\}$ is necessary in our study. Especially, in the incidence matrix $T_L^\leq = (t_{ij})$ we have two cases:

- (i) $t_{ij} = 1$, when $x_i \leq x_j$ or
- (ii) $t_{ij} = 0$, otherwise.

Therefore, if we have $t_{ij} = 1$, then either $x_i < x_j$ or $x_i = x_j$. Also, if we have $t_{ij} = 0$, then $x_j < x_i$ or $x_j \parallel x_i$.

Undoubtedly, the incidence matrices play an essential role in Topology and related studies. Using this matrix we can have different characterizations of various notions like the height of a poset (see (Dube et al. 2017)) and various relations (see the following Proposition 1). However, the necessity to be more specific with the relation between two elements of a lattice leads to the so-called order matrix $A_L^\leq = (a_{ij})$. This matrix contains all the cases that we can have between any two elements x_i, x_j of the lattice L :

- (i) $x_i < x_j$,
- (ii) $x_j < x_i$,
- (iii) $x_i = x_j$ and
- (iv) $x_i \parallel x_j$.

This detailed separation of all possible relations is the reason for the use of this matrix. We can use this matrix in order to succeed more results in Order Theory, enriching related studies. For example, if $a_{ij} = 1$, then we are sure that $x_i = x_j$ and if $a_{ij} = 2$, then certainly, $x_i < x_j$. Thus, the combination of these two matrices gives us the desired results and characterizations for the concepts of the paper.

In the following Proposition 1 we also use the $n \times 1$ matrix $B_1 = (b_1^i)$, where

$$b_1^i = \begin{cases} 1, & \text{if } i = 1 \\ 0, & \text{otherwise.} \end{cases}$$

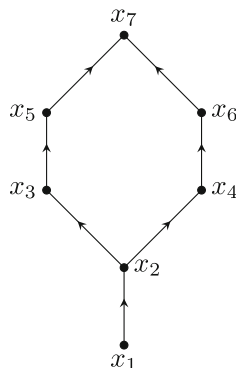
Proposition 1 *Let $(L, \leq)$ be a finite lattice. For each $x_i \in L$, $i \in \{1, \dots, n\}$, an element $x_j \in L$, $j \in \{1, 2, \dots, n\} \setminus \{i\}$ satisfies the relation $x_i \wedge x_j = x_1$, where $x_1 = 0_L$, if and only if*

$$\max(C_i(T_L^\leq) + C_j(T_L^\leq) - B_1) = 1.$$

Proof Let $x_i \in L$, $i \in \{1, 2, \dots, n\}$, $x_j \in L$, $j \in \{1, 2, \dots, n\} \setminus \{i\}$ such that $x_i \wedge x_j = x_1$ and

$$\max(C_i(T_L^\leq) + C_j(T_L^\leq) - B_1) = k.$$

Then we have that $k = 1$ or $k = 2$. Since $x_i \wedge x_j = x_1$, we have that $x_1 \leq x_i$ and $x_1 \leq x_j$. Thus, $t_{1i} = t_{1j} = 1$. Additionally, $b_1^1 = 1$. Therefore, $t_{1i} + t_{1j} - b_1^1 = 1 + 1 - 1 = 1$ and thus, $k = 1$. (We also state that if there exists $i_0 \in \{2, \dots, n\}$ such that $t_{i_0i} + t_{i_0j} - b_1^{i_0} = 2$, then $t_{i_0i} = t_{i_0j} = 1$ and $b_1^{i_0} = 0$. Thus, $x_{i_0} \leq x_i$ and $x_{i_0} \leq x_j$ and therefore, $x_{i_0} = x_1$, which is a contradiction.)

Fig. 1 The finite lattice $(L, \leq)$ 

Conversely, let $x_i \in L, i \in \{1, 2, \dots, n\}$ and

$$\max(C_i(T_L^{\leq}) + C_j(T_L^{\leq}) - B_1) = 1,$$

for every $j \in \{1, 2, \dots, n\} \setminus \{i\}$. We shall prove that $x_i \wedge x_j = x_1$, for every $j \in \{1, 2, \dots, n\} \setminus \{i\}$. In contrast, we suppose that there exists $j_0 \in \{1, 2, \dots, n\} \setminus \{i\}$ such that $x_i \wedge x_{j_0} \neq x_1$. We suppose that $x_i \wedge x_{j_0} = x_m$. Then $t_{mi} = t_{mj_0} = 1$ and $b_1^m = 0$. Thus, $t_{mi} + t_{mj} - b_1^m = 1 + 1 - 0 = 2$ and therefore,

$$\max(C_i(T_L^{\leq}) + C_{j_0}(T_L^{\leq}) - B_1) = 2,$$

which is a contradiction. Thus, $x_i \wedge x_j = x_1$, for every $j \in \{1, 2, \dots, n\} \setminus \{i\}$. $\square$

Example 1 We consider the lattice $(L, \leq)$ which is represented in Fig. 1.

The incidence matrix of L is the following 7×7 matrix

$$T_L^{\leq} = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}.$$

For the element x_3 we have

$$C_3(T_L^{\leq}) + C_1(T_L^{\leq}) - B_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} - \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

and

$$\max(C_3(T_L^{\leq}) + C_1(T_L^{\leq}) - B_1) = 1.$$

Also, for every $j \in \{2, 4, 5, 6, 7\}$ we have that

$$\max(C_3(T_L^{\leq}) + C_j(T_L^{\leq}) - B_1) = 2.$$

Thus, x_1 is the only element of L for which $x_3 \wedge x_1 = x_1$.

Proposition 2 *Let $(L, \leq)$ be a finite lattice and $\{x_{j_1}, \dots, x_{j_m}\}$ be a set consisting of distinct elements of L . Then an element x_{i_0} of L is an upper bound of the set $\{x_{j_1}, \dots, x_{j_m}\}$ if and only if one of the following conditions is satisfied:*

$$2m + 1 \in r_{j_1}(\mathbf{A}_L^{\leq}) + \dots + r_{j_m}(\mathbf{A}_L^{\leq}) + r_{i_0}(\mathbf{A}_L^{\leq}),$$

if $\{x_{j_1}, \dots, x_{j_m}\}$ is not a chain and $i_0 \notin \{j_1, \dots, j_m\}$,
or

$$2m + 1 \notin r_{j_1}(\mathbf{A}_L^{\leq}) + \dots + r_{j_m}(\mathbf{A}_L^{\leq}) + r_{i_0}(\mathbf{A}_L^{\leq}),$$

if $\{x_{j_1}, \dots, x_{j_m}\}$ is a chain and $i_0 \in \{j_1, \dots, j_m\}$.

Proof We separate the proof in the following two cases:

Case 1: If x_{i_0} is an upper bound of the set $\{x_{j_1}, \dots, x_{j_m}\}$ such that $\{x_{j_1}, \dots, x_{j_m}\}$ is not a chain with $i_0 \notin \{j_1, \dots, j_m\}$, then $x_{j_1} < x_{i_0}, \dots, x_{j_m} < x_{i_0}$. Thus, $a_{j_1 i_0} = \dots = a_{j_m i_0} = 2$ and since $a_{i_0 i_0} = 1$, we have that

$$2m + 1 \in r_{j_1}(\mathbf{A}_L^{\leq}) + \dots + r_{j_m}(\mathbf{A}_L^{\leq}) + r_{i_0}(\mathbf{A}_L^{\leq}).$$

Conversely, we suppose that $2m + 1 \in r_{j_1}(\mathbf{A}_L^{\leq}) + \dots + r_{j_m}(\mathbf{A}_L^{\leq}) + r_{i_0}(\mathbf{A}_L^{\leq})$, with $\{x_{j_1}, \dots, x_{j_m}\}$ is not a chain and $i_0 \notin \{j_1, \dots, j_m\}$. Then there exists $k \in \{1, \dots, n\}$ such that $a_{j_1 k} = \dots = a_{j_m k} = 2$ and $a_{i_0 k} = 1$. Thus, x_k is an upper bound of the set $\{x_{j_1}, \dots, x_{j_m}\}$ (since $x_{j_1} < x_k, \dots, x_{j_m} < x_k$) and $x_{i_0} = x_k$. Therefore, x_{i_0} is an upper bound of the set $\{x_{j_1}, \dots, x_{j_m}\}$.

Case 2: We suppose that x_{i_0} is an upper bound of the set $\{x_{j_1}, \dots, x_{j_m}\}$ such that $\{x_{j_1}, \dots, x_{j_m}\}$ is a chain with $i_0 \in \{j_1, \dots, j_m\}$. Let $x_{i_0} = x_{j_r}$, where $r \in \{1, \dots, m\}$. In contrast, we suppose that

$$2m + 1 \in r_{j_1}(\mathbf{A}_L^{\leq}) + \dots + r_{j_m}(\mathbf{A}_L^{\leq}) + r_{i_0}(\mathbf{A}_L^{\leq}).$$

Then there exists $k \in \{1, \dots, n\}$ such that $a_{j_1 k} = \dots = a_{j_m k} = 2$ and $a_{i_0 k} = 1$. Thus, $a_{j_r k} = a_{i_0 k} = 2$ and $a_{i_0 k} = 1$, which is a contradiction.

Conversely, we suppose that $2m + 1 \notin r_{j_1}(\mathbf{A}_L^{\leq}) + \dots + r_{j_m}(\mathbf{A}_L^{\leq}) + r_{i_0}(\mathbf{A}_L^{\leq})$, with $\{x_{j_1}, \dots, x_{j_m}\}$ is a chain and $i_0 \in \{j_1, \dots, j_m\}$. Similarly, let $x_{i_0} = x_{j_r}$, where $r \in \{1, \dots, m\}$. In contrast, we suppose that x_{i_0} is not an upper bound of $\{x_{j_1}, \dots, x_{j_m}\}$. Since $\{x_{j_1}, \dots, x_{j_m}\}$ is a chain, there exists an element $x_k \in \{x_{j_1}, \dots, x_{j_m}\}$, different from x_{i_0} , which is an upper bound of $\{x_{j_1}, \dots, x_{j_m}\}$. Let $x_k = x_{j_l}$, where $l \in \{1, \dots, m\} \setminus \{r\}$. Then

$$\underbrace{a_{j_1 k} = \dots = a_{j_{l-1} k} = a_{j_{l+1} k} = \dots = a_{j_m k} = 2}_{(m-1)\text{-times}}$$

$a_{j_l j_l} = 1$ and $a_{i_0 k} = 2$. Therefore,

$$2(m-1) + 1 + 2 \in r_{j_1}(\mathbf{A}_L^{\leq}) + \dots + r_{j_m}(\mathbf{A}_L^{\leq}) + r_{i_0}(\mathbf{A}_L^{\leq})$$

and equivalently,

$$2m + 1 \in r_{j_1}(\mathbf{A}_L^{\leq}) + \dots + r_{j_m}(\mathbf{A}_L^{\leq}) + r_{i_0}(\mathbf{A}_L^{\leq}),$$

which is a contradiction. □

Proposition 3 Let $(L, \leq)$ be a finite lattice and $\{x_{j_1}, \dots, x_{j_m}\}$ be a set consisting of distinct elements of L . An upper bound x_{i_0} of the set $\{x_{j_1}, \dots, x_{j_m}\}$ is the least upper bound of it if and only if one of the following conditions is satisfied:

$$2m + 2 \notin r_{j_1}(\mathbf{A}_L^{\leq}) + \dots + r_{j_m}(\mathbf{A}_L^{\leq}) + (-1) \cdot r_{i_0}(\mathbf{A}_L^{\leq}), \quad (\text{LU1})$$

if $\{x_{j_1}, \dots, x_{j_m}\}$ is not a chain and $i_0 \notin \{j_1, \dots, j_m\}$,
or

$$2m - 1 \notin r_{j_1}(\mathbf{A}_L^{\leq}) + \dots + r_{j_m}(\mathbf{A}_L^{\leq}) + (-1) \cdot r_{i_0}(\mathbf{A}_L^{\leq}), \quad (\text{LU2})$$

if $\{x_{j_1}, \dots, x_{j_m}\}$ is a chain and $i_0 \in \{j_1, \dots, j_m\}$.

Proof We separate the proof in the following two cases:

Case 1: We suppose that $x_{i_0} = x_{j_1} \vee \dots \vee x_{j_m}$ such that $\{x_{j_1}, \dots, x_{j_m}\}$ is not a chain and $i_0 \notin \{j_1, \dots, j_m\}$ with

$$2m + 2 \in r_{j_1}(\mathbf{A}_L^{\leq}) + \dots + r_{j_m}(\mathbf{A}_L^{\leq}) + (-1) \cdot r_{i_0}(\mathbf{A}_L^{\leq}).$$

Then there exists $k \in \{1, \dots, n\}$ such that $a_{j_1 k} = \dots = a_{j_m k} = 2$ and $a_{i_0 k} = -2$. Thus, x_k is an upper bound of $\{x_{j_1}, \dots, x_{j_m}\}$ with $x_k < x_{i_0}$, which is a contradiction.

Conversely, we suppose that

$$2m + 2 \notin r_{j_1}(\mathbf{A}_L^{\leq}) + \dots + r_{j_m}(\mathbf{A}_L^{\leq}) + (-1) \cdot r_{i_0}(\mathbf{A}_L^{\leq})$$

and prove that x_{i_0} is the least upper bound of the set $\{x_{j_1}, \dots, x_{j_m}\}$, where $\{x_{j_1}, \dots, x_{j_m}\}$ is not a chain and $i_0 \notin \{j_1, \dots, j_m\}$. In contrast, we suppose that $x_{j_1} \vee \dots \vee x_{j_m} = x_k$ with $x_k < x_{i_0}$. Then $a_{j_1 k} = \dots = a_{j_m k} = 2$ and $a_{i_0 k} = -2$. Therefore,

$$2m + 2 \in r_{j_1}(\mathbf{A}_L^{\leq}) + \dots + r_{j_m}(\mathbf{A}_L^{\leq}) + (-1) \cdot r_{i_0}(\mathbf{A}_L^{\leq}),$$

which is a contradiction.

Case 2: We suppose that x_{i_0} is the least upper bound of the set $\{x_{j_1}, \dots, x_{j_m}\}$ such that $\{x_{j_1}, \dots, x_{j_m}\}$ is a chain and $i_0 \in \{j_1, \dots, j_m\}$. We suppose that $x_{i_0} = x_{j_r}$, where $r \in \{1, \dots, m\}$. In contrast, we suppose that

$$2m - 1 \in r_{j_1}(\mathbf{A}_L^{\leq}) + \dots + r_{j_m}(\mathbf{A}_L^{\leq}) + (-1) \cdot r_{i_0}(\mathbf{A}_L^{\leq}).$$

Then there exists $k \in \{1, \dots, n\}$ such that $a_{j_1 k} = \dots = a_{j_m k} = 2$ and $a_{i_0 k} = 1$. Thus, $a_{j_r k} = a_{i_0 k} = 2$ and $a_{i_0 k} = 1$, which is a contradiction.

Conversely, we suppose that

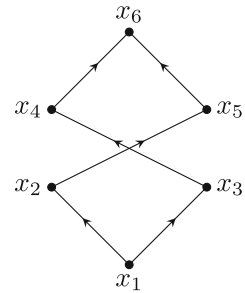
$$2m - 1 \notin r_{j_1}(\mathbf{A}_L^{\leq}) + \dots + r_{j_m}(\mathbf{A}_L^{\leq}) + (-1) \cdot r_{i_0}(\mathbf{A}_L^{\leq}),$$

with $\{x_{j_1}, \dots, x_{j_m}\}$ is a chain and $i_0 \in \{j_1, \dots, j_m\}$. Similarly, let $x_{i_0} = x_{j_r}$, where $r \in \{1, \dots, m\}$. In contrast, we suppose that $x_{j_1} \vee \dots \vee x_{j_m} = x_k$ with $x_k < x_{i_0}$. Then $a_{j_1 k} = \dots = a_{j_m k} = 2$ and $a_{i_0 k} = -2$. Therefore, $a_{i_0 k} = 2$ and $a_{i_0 k} = -2$, which is a contradiction. $\square$

Theorem 4 Let $(L, \leq)$ be a finite lattice, $x_i \in L$, where $i \in \{1, \dots, n\}$, and $\{x_{j_1}, \dots, x_{j_m}\}$ be the set which consists of distinct elements of L for which

$$\max(C_i(T_L^{\leq}) + C_{j_s}(T_L^{\leq}) - B_1) = 1,$$

for every $s \in \{1, \dots, m\}$. Then an element x_{i_0} of L is the pseudocomplement of x_i if and only if it satisfies the following conditions:

Fig. 2 The finite lattice $(L, \leq)$ 

(U) $(U1) 2m + 1 \in r_{j_1}(A_L^{\leq}) + \dots + r_{j_m}(A_L^{\leq}) + r_{i_0}(A_L^{\leq})$, if $\{x_{j_1}, \dots, x_{j_m}\}$ is not a chain and $i_0 \notin \{j_1, \dots, j_m\}$ or

$(U2) 2m + 1 \notin r_{j_1}(A_L^{\leq}) + \dots + r_{j_m}(A_L^{\leq}) + r_{i_0}(A_L^{\leq})$, if $\{x_{j_1}, \dots, x_{j_m}\}$ is a chain and $i_0 \in \{j_1, \dots, j_m\}$

and

(LU) $(LU1) 2m + 2 \notin r_{j_1}(A_L^{\leq}) + \dots + r_{j_m}(A_L^{\leq}) + (-1) \cdot r_{i_0}(A_L^{\leq})$, if $\{x_{j_1}, \dots, x_{j_m}\}$ is not a chain and $i_0 \notin \{j_1, \dots, j_m\}$ or

$(LU2) 2m - 1 \notin r_{j_1}(A_L^{\leq}) + \dots + r_{j_m}(A_L^{\leq}) + (-1) \cdot r_{i_0}(A_L^{\leq})$, if $\{x_{j_1}, \dots, x_{j_m}\}$ is a chain and $i_0 \in \{j_1, \dots, j_m\}$.

Proof The pseudocomplement of $x_i \in L$, where $i \in \{1, \dots, n\}$, is defined as follows:

$$x_i^* = x_{i_0} = \bigvee \{x_j \in L : x_i \wedge x_j = x_1\},$$

that is x_i^* is the least upper bound of the set $\{x_j \in L : x_i \wedge x_j = x_1\}$.

Let $\{x_{j_1}, \dots, x_{j_m}\}$ be the set which consists of distinct elements of L for which

$$\max(C_i(T_L^{\leq}) + C_{j_s}(T_L^{\leq}) - B_1) = 1,$$

for every $s \in \{1, \dots, m\}$. By Proposition 1, these elements $x_{j_1}, \dots, x_{j_m}$ are those for which $x_i \wedge x_{j_s} = x_1$, $s \in \{1, \dots, m\}$. Therefore, $x_i^* = x_{i_0}$ is the least upper bound of the set $\{x_{j_1}, \dots, x_{j_m}\}$ for which

$$\max(C_i(T_L^{\leq}) + C_{j_s}(T_L^{\leq}) - B_1) = 1,$$

for every $s \in \{1, \dots, m\}$.

If x_{i_0} is the pseudocomplement of x_i , then it is an upper bound of the set $\{x_{j_1}, \dots, x_{j_m}\}$ and thus, by Proposition 2, condition (U) is satisfied. Moreover, x_{i_0} is the least upper bound of the set $\{x_{j_1}, \dots, x_{j_m}\}$ and thus, by Proposition 3 condition (LU) is satisfied.

Conversely, if x_{i_0} is an element of L which satisfies the condition (U), then by Proposition 2, x_{i_0} is an upper bound of $\{x_{j_1}, \dots, x_{j_m}\}$. Also, if x_{i_0} satisfies the condition (LU), then by Proposition 3, x_{i_0} is the least upper bound of $\{x_{j_1}, \dots, x_{j_m}\}$. Therefore, x_{i_0} is the pseudocomplement of x_i . $\square$

Example 2 We consider the lattice $(L, \leq)$ which is represented in Fig. 2.

For example, we study the element x_2 . Following Proposition 1 we see that the elements x_1, x_3 and x_4 are those for which $x_2 \wedge x_1 = x_1$, $x_2 \wedge x_3 = x_1$ and $x_2 \wedge x_4 = x_1$. Then we study the set $\{x_1, x_3, x_4\}$, which is a chain and therefore, an upper bound of it is one of the elements x_1, x_3 or x_4 .

The order matrix of L is the following 6×6 matrix

$$\mathbf{A}_L^{\leq} = \begin{pmatrix} 1 & 2 & 2 & 2 & 2 & 2 \\ -2 & 1 & 0 & 0 & 2 & 2 \\ -2 & 0 & 1 & 2 & 0 & 2 \\ -2 & 0 & -2 & 1 & 0 & 2 \\ -2 & -2 & 0 & 0 & 1 & 2 \\ -2 & -2 & -2 & -2 & -2 & 1 \end{pmatrix}.$$

By Proposition 2 we have

$$2 \cdot 3 + 1 \in r_1(\mathbf{A}_L^{\leq}) + r_3(\mathbf{A}_L^{\leq}) + r_4(\mathbf{A}_L^{\leq}) + r_1(\mathbf{A}_L^{\leq}),$$

and thus, x_1 is not an upper bound of the set $\{x_1, x_3, x_4\}$. Also,

$$2 \cdot 3 + 1 \in r_1(\mathbf{A}_L^{\leq}) + r_3(\mathbf{A}_L^{\leq}) + r_4(\mathbf{A}_L^{\leq}) + r_3(\mathbf{A}_L^{\leq}),$$

and thus, x_3 is not an upper bound of the set $\{x_1, x_3, x_4\}$. Finally,

$$2 \cdot 3 + 1 \notin r_1(\mathbf{A}_L^{\leq}) + r_3(\mathbf{A}_L^{\leq}) + r_4(\mathbf{A}_L^{\leq}) + r_1(\mathbf{A}_L^{\leq}),$$

and thus, x_4 is an upper bound of the set $\{x_1, x_3, x_4\}$.

Also, by Proposition 3, since

$$2 \cdot 3 - 1 \notin r_1(\mathbf{A}_L^{\leq}) + r_3(\mathbf{A}_L^{\leq}) + r_4(\mathbf{A}_L^{\leq}) + (-1) \cdot r_4(\mathbf{A}_L^{\leq}),$$

x_4 is the least upper bound of the set $\{x_1, x_3, x_4\}$.

Therefore, by Theorem 4 we have that $x_2^* = x_4$.

The facts of Propositions 1, 2, 3 and Theorem 4 can be presented in a matrix algorithmic procedure which can find the pseudocomplement of an arbitrary element x_i , $i \in \{1, \dots, n\}$, of a finite lattice $L = \{x_1, \dots, x_n\}$.

We state that this algorithm computes the pseudocomplement of each element of L , individually. This idea is based on the fact that in order to compute the small inductive dimension (see the corresponding definition of it in Sect. 1) we have to study one-by-one the elements and their pseudocomplements of covers. Thus, as we will see in Theorem 8, it is more clear to study the pseudocomplement of certain elements, separately.

Especially, if $L = \{x_1, \dots, x_n\}$ is a finite lattice with $n \geq 2$ and x_i , $i \in \{1, \dots, n\}$, is an arbitrary element of L , then this procedure is given as follows, named as Algorithm 1.

1. Find the incidence matrix $T_L^{\leq}$ and the order matrix $\mathbf{A}_L^{\leq}$ of L and go to Step 2.
2. Set $j = 1$, with $j \neq i$, $m = 1$ and go to Step 3.
3. Find the matrix $C_i(T_L^{\leq}) + C_j(T_L^{\leq}) - B_1$. If

$$\max(C_i(T_L^{\leq}) + C_j(T_L^{\leq}) - B_1) = 1,$$

then put $x_j = x_{j_m}$ and go to Step 4. Otherwise, go also to Step 4.

4. If $j < n$, then put $j \rightarrow j + 1$ (that is, replace the old j with $j + 1$) and go to Step 3. Otherwise, go to Step 5.
5. Print the set $\{x_{j_1}, \dots, x_{j_m}\}$ and go to Step 6.
6. Set $l = 1$ and go to Step 7.
7. If $\{x_{j_1}, \dots, x_{j_m}\}$ is not a chain with $l \notin \{j_1, \dots, j_m\}$ and

$$2m + 1 \in r_{j_1}(\mathbf{A}_L^{\leq}) + \dots + r_{j_m}(\mathbf{A}_L^{\leq}) + r_l(\mathbf{A}_L^{\leq})$$

or

if $\{x_{j_1}, \dots, x_{j_m}\}$ is a chain with $l \in \{j_1, \dots, j_m\}$ and

$$2m + 1 \notin r_{j_1}(\mathbf{A}_L^{\leq}) + \dots + r_{j_m}(\mathbf{A}_L^{\leq}) + r_l(\mathbf{A}_L^{\leq}),$$

then go to Step 9. Otherwise, go to Step 8.

8. If $l < n$, then put $l \rightarrow l + 1$ (that is, replace the old l with $l + 1$) and go to Step 7. Otherwise, go to Step 10.
9. If $\{x_{j_1}, \dots, x_{j_m}\}$ is not a chain with $l \notin \{j_1, \dots, j_m\}$ and

$$2m + 2 \notin r_{j_1}(\mathbf{A}_L^{\leq}) + \dots + r_{j_m}(\mathbf{A}_L^{\leq}) + (-1) \cdot r_l(\mathbf{A}_L^{\leq})$$

or

if $\{x_{j_1}, \dots, x_{j_m}\}$ is a chain with $l \in \{j_1, \dots, j_m\}$ and

$$2m - 1 \notin r_{j_1}(\mathbf{A}_L^{\leq}) + \dots + r_{j_m}(\mathbf{A}_L^{\leq}) + (-1) \cdot r_l(\mathbf{A}_L^{\leq}),$$

then print $x_i^* = x_l$. Otherwise, go to Step 8.

10. Print $x_i^* = x_n$.

3 The lattice $\uparrow(x \vee x^*)$ with matrices

According to the definition of the small inductive dimension, it is considered to be necessary to study also the lattice $\uparrow(x_i \vee x_i^*)$, where $i \in \{1, \dots, n\}$. Therefore, in this section, based on the results of Sect. 2, we study the element $x_i \vee x_i^*$, where $i \in \{1, \dots, n\}$, using order matrices and then we study the corresponding finite lattice $\uparrow(x_i \vee x_i^*)$, through these matrices. An algorithm, which collects all these results and provides a way to determine this lattice using matrices, is presented also in this section.

Proposition 5 *Let $(L, \leq)$ be a finite lattice, $x_i \in L$, where $i \in \{1, \dots, n\}$, and $x_{i_1} = x_i^*$ the pseudocomplement of x_i . Then an element x_{i_0} of L is an upper bound of the set $\{x_i, x_{i_1}\}$ if and only if one of the following conditions is satisfied:*

$$5 \in r_i(\mathbf{A}_L^{\leq}) + r_{i_1}(\mathbf{A}_L^{\leq}) + r_{i_0}(\mathbf{A}_L^{\leq}), \quad (TU1)$$

if $\{x_i, x_{i_1}\}$ is an antichain and $i_0 \notin \{i, i_1\}$

or

$$5 \notin r_i(\mathbf{A}_L^{\leq}) + r_{i_1}(\mathbf{A}_L^{\leq}) + r_{i_0}(\mathbf{A}_L^{\leq}), \quad (TU2)$$

if $\{x_i, x_{i_1}\}$ is a chain and $i_0 \in \{i, i_1\}$.

Proof Since the set $\{x_i, x_{i_1}\}$, that we want to study, consists of two elements, by Proposition 2 for $m = 2$ we have that an element x_{i_0} of L is an upper bound of $\{x_i, x_{i_1}\}$ if and only if one of the following conditions is satisfied:

$$5 = 2 \cdot 2 + 1 \in r_i(\mathbf{A}_L^{\leq}) + r_{i_1}(\mathbf{A}_L^{\leq}) + r_{i_0}(\mathbf{A}_L^{\leq}), \quad (TU1)$$

if $\{x_i, x_{i_1}\}$ is not a chain (and therefore is an antichain) and $i_0 \notin \{i, i_1\}$

or

$$5 = 2 \cdot 2 + 1 \notin r_i(\mathbf{A}_L^{\leq}) + r_{i_1}(\mathbf{A}_L^{\leq}) + r_{i_0}(\mathbf{A}_L^{\leq}) \quad (TU2)$$

if $\{x_i, x_{i_1}\}$ is a chain and $i_0 \in \{i, i_1\}$. □

Proposition 6 Let $(L, \leq)$ be a finite lattice, $x_i \in L$, where $i \in \{1, \dots, n\}$, and $x_{i_1} = x_i^*$ the pseudocomplement of x_i . An upper bound x_{i_0} of the set $\{x_i, x_{i_1}\}$ is the least upper bound of it if and only if one of the following conditions is satisfied:

$$6 \notin r_i(\mathbf{A}_L^{\leq}) + r_{i_1}(\mathbf{A}_L^{\leq}) + (-1) \cdot r_{i_0}(\mathbf{A}_L^{\leq}), \quad (LTU1)$$

if $\{x_i, x_{i_1}\}$ is an antichain and $i_0 \notin \{i, i_1\}$
or

$$3 \notin r_i(\mathbf{A}_L^{\leq}) + r_{i_1}(\mathbf{A}_L^{\leq}) + (-1) \cdot r_{i_0}(\mathbf{A}_L^{\leq}), \quad (LTU2)$$

if $\{x_i, x_{i_1}\}$ is a chain and $i_0 \in \{i, i_1\}$.

Proof Since the set $\{x_i, x_{i_1}\}$, that we want to study, consists of two elements, by Proposition 3 for $m = 2$ we have that an upper bound x_{i_0} of the set $\{x_i, x_{i_1}\}$ is the least upper bound of it if and only if one of the following conditions is satisfied:

$$6 = 2 \cdot 2 + 2 \notin r_i(\mathbf{A}_L^{\leq}) + r_{i_1}(\mathbf{A}_L^{\leq}) + (-1) \cdot r_{i_0}(\mathbf{A}_L^{\leq}), \quad (LTU1)$$

if $\{x_i, x_{i_1}\}$ is not a chain (and therefore is an antichain) and $i_0 \notin \{i, i_1\}$
or

$$3 = 2 \cdot 2 - 1 \notin r_i(\mathbf{A}_L^{\leq}) + r_{i_1}(\mathbf{A}_L^{\leq}) + (-1) \cdot r_{i_0}(\mathbf{A}_L^{\leq}), \quad (LTU2)$$

if $\{x_i, x_{i_1}\}$ is a chain and $i_0 \in \{i, i_1\}$. $\square$

Proposition 7 Let $(L, \leq)$ be a finite lattice, $\mathbf{A}_L^{\leq} = (a_{ij})$ its order matrix and $x_i \in L$, where $i \in \{1, \dots, n\}$. Then

$$\uparrow x_i = \{x_j \in L : a_{ij} \in \{1, 2\}\}.$$

Proof For the element $x_i \in L$, $i \in \{1, \dots, n\}$, we have

$$\uparrow x_i = \{x_j \in L : x_i \leq x_j\}.$$

According to the definition of the order matrix $\mathbf{A}_L^{\leq} = (a_{ij})$ of L , we have that

$$x_i \leq x_j \Leftrightarrow \begin{cases} a_{ij} = 1, & \text{if } x_i = x_j, \\ a_{ij} = 2, & \text{if } x_i < x_j. \end{cases}$$

Therefore,

$$\uparrow x_i = \{x_j \in L : a_{ij} \in \{1, 2\}\}.$$

$\square$

Example 3 We consider the lattice $(L, \leq)$ of Fig. 2. We have seen that the pseudocomplement of x_2 is the element x_4 . Since the set $\{x_2, x_4\}$ is an antichain, an upper bound of it is one of the elements x_1, x_3, x_5, x_6 .

By Proposition 5, since

$$5 \notin r_2(\mathbf{A}_L^{\leq}) + r_4(\mathbf{A}_L^{\leq}) + r_1(\mathbf{A}_L^{\leq}),$$

x_1 is not an upper bound of the set $\{x_2, x_4\}$. Moreover, since

$$5 \notin r_2(\mathbf{A}_L^{\leq}) + r_4(\mathbf{A}_L^{\leq}) + r_3(\mathbf{A}_L^{\leq}),$$

x_3 is not an upper bound of the set $\{x_2, x_4\}$. Since

$$5 \notin r_2(\mathbf{A}_L^{\leq}) + r_4(\mathbf{A}_L^{\leq}) + r_5(\mathbf{A}_L^{\leq}),$$

x_5 is not an upper bound of the set $\{x_2, x_4\}$. Finally, since

$$5 \in r_2(\mathbf{A}_L^{\leq}) + r_4(\mathbf{A}_L^{\leq}) + r_6(\mathbf{A}_L^{\leq}),$$

x_6 is an upper bound of the set $\{x_2, x_4\}$.

In addition, by Proposition 6, since

$$6 \notin r_2(\mathbf{A}_L^{\leq}) + r_4(\mathbf{A}_L^{\leq}) + (-1) \cdot r_6(\mathbf{A}_L^{\leq}),$$

x_6 is the least upper bound of the set $\{x_2, x_4\}$.

By Proposition 7 we have that

$$\uparrow x_6 = \{x_j : a_{6j} \in \{1, 2\}\} = \{x_6\}.$$

Based on the facts of Propositions 5, 6 and 7, we present an algorithm which determines the lattice $\uparrow(x \vee x^*)$ through incidence and order matrices. Similar to Algorithm 1, it is more familiar to study each element x of L , individually. Thus, we can apply it only for those elements of minimal covers (see Theorem 8) that are needed for the small inductive dimension.

Especially, if $L = \{x_1, \dots, x_n\}$ is a finite lattice with $n \geq 2$ and $x_i \in L$, $i \in \{1, \dots, n\}$, then this procedure is given as follows, named as Algorithm 2.

1. Apply Algorithm 1 in order to find the pseudocomplement x_{i_1} of x_i and go to Step 2.
2. Set $l = 1$ and go to Step 3.
3. If $\{x_i, x_{i_1}\}$ is an antichain with $l \notin \{i, i_1\}$ and

$$5 \in r_i(\mathbf{A}_L^{\leq}) + r_{i_1}(\mathbf{A}_L^{\leq}) + r_l(\mathbf{A}_L^{\leq}),$$

or

if $\{x_i, x_{i_1}\}$ is a chain with $l \notin \{i, i_1\}$ and

$$5 \notin r_i(\mathbf{A}_L^{\leq}) + r_{i_1}(\mathbf{A}_L^{\leq}) + r_l(\mathbf{A}_L^{\leq}),$$

then go to Step 5. Otherwise, go to Step 4.

4. If $l < n$, then put $l \rightarrow l + 1$ (that is, replace the old l with $l + 1$) and go to Step 3. Otherwise, go to Step 7.
5. If $\{x_i, x_{i_1}\}$ is an antichain with $l \notin \{i, i_1\}$ and

$$6 \notin r_i(\mathbf{A}_L^{\leq}) + r_{i_1}(\mathbf{A}_L^{\leq}) + (-1) \cdot r_l(\mathbf{A}_L^{\leq})$$

or

if $\{x_i, x_{i_1}\}$ is a chain with $l \in \{i, i_1\}$ and

$$3 \notin r_i(\mathbf{A}_L^{\leq}) + r_{i_1}(\mathbf{A}_L^{\leq}) + (-1) \cdot r_l(\mathbf{A}_L^{\leq}),$$

then print $x_i \vee x_{i_1} = x_l$ and go to Step 6. Otherwise, go to Step 4.

6. Find the elements $x_t \in L$ for which $a_{lt} \in \{1, 2\}$ and print

$$\uparrow(x_i \vee x_{i_1}) = \uparrow x_l = \{x_t \in L : a_{lt} \in \{1, 2\}\}.$$

7. Print $\uparrow(x_i \vee x_{i_1}) = \uparrow x_n = \{x_n\}$.

4 Algorithmic procedure of the small inductive dimension for finite lattices

In this section, we present the connection of the small inductive dimension with the so-called minimal covers. A combination of Algorithms 1 and 2 of the previous sections leads us to present an algorithm for the computation of the small inductive dimension of an arbitrary finite lattice using matrices.

Theorem 8 *If $(L, \leq)$ is a finite lattice, then we have that $\text{ind}(L) \leq k$, where $k \in \{0, 1, \dots\}$, if and only if for every minimal cover V of L , we have*

$$\text{ind}(\uparrow(v^* \vee v)) \leq k - 1,$$

for every $v \in V$.

Proof Let $\text{ind}(L) \leq k$, where $k \in \{0, 1, \dots\}$, V be a minimal cover of L and $v \in V$. We shall prove that $\text{ind}(\uparrow(v^* \vee v)) \leq k - 1$. Since $\text{ind}(L) \leq k$, there exists a cover U of L , which refines V and for every $u \in U$ we have that

$$\text{ind}(\uparrow(u^* \vee u)) \leq k - 1.$$

Since V is a minimal cover of L , we have that $V \subseteq U$ and thus, $v \in U$. Therefore, $\text{ind}(\uparrow(v^* \vee v)) \leq k - 1$.

Conversely, we suppose that for every minimal cover V of L , we have

$$\text{ind}(\uparrow(v^* \vee v)) \leq k - 1,$$

for every $v \in V$, where $k \in \{0, 1, \dots\}$, and we shall prove that $\text{ind}(L) \leq k$. Let U be a cover of L . We shall find a cover W of L , which refines U and $\text{ind}(\uparrow(w^* \vee w)) \leq k - 1$, for every $w \in W$. For that, we study the following cases:

(i) If $1_L \in U$, then the set $W = \{1_L\}$ is a cover of L which refines U and

$$\text{ind}(\uparrow(1_L^* \vee 1_L)) = \text{ind}(\uparrow(0_L \vee 1_L)) = \text{ind}(\uparrow 1_L) = \text{ind}(\{1_L\}) = -1 \leq k - 1.$$

(ii) If $1_L \notin U$, then by (Dube et al. (2015), Lemma 3.7), there exists a minimal cover W of L which refines U and $1_L \notin W$. By assumption, since W is a minimal cover of L , we have that $\text{ind}(\uparrow(w^* \vee w)) \leq k - 1$, for every $w \in W$.

Therefore, in each of the above two cases we have that $\text{ind}(L) \leq k$. $\square$

Proposition 9 *If $(L, \leq)$ is a finite lattice, then we have that*

$$\text{ind}(L) = \max\{\text{ind}(\uparrow(v \vee v^*)) + 1 : v \in V, V \text{ minimal cover of } L\}.$$

Proof Let $k \in \{0, 1, \dots\}$ and

$$\max\{\text{ind}(\uparrow(v \vee v^*)) + 1 : v \in V, V \text{ minimal cover of } L\} = k.$$

We shall prove that $\text{ind}(L) \leq k$. We have that $\text{ind}(\uparrow(v \vee v^*)) + 1 \leq k$, for every minimal cover V of L and for every $v \in V$. Then $\text{ind}(\uparrow(v \vee v^*)) \leq k - 1$, for every minimal cover V of L and for every $v \in V$ and thus, by Theorem 8, $\text{ind}(L) \leq k$.

Conversely, we suppose that $\text{ind}(L) = k$ and we prove that

$$\max\{\text{ind}(\uparrow(v \vee v^*)) + 1 : v \in V, V \text{ minimal cover of } L\} \leq k.$$

Since $\text{ind}(L) = k$, by Theorem 8, for every minimal cover V of L and for every $v \in V$ we have that $\text{ind}(\uparrow(v \vee v^*)) \leq k - 1$ or equivalently, for every minimal cover V of L and for every $v \in V$ we have that $\text{ind}(\uparrow(v \vee v^*)) + 1 \leq k$. Thus,

$$\max\{\text{ind}(\uparrow(v \vee v^*)) + 1 : v \in V, V \text{ minimal cover of } L\} \leq k.$$

□

Proposition 10 *If $0 \notin A_L^{\leq}$, then $\text{ind}(L) = 0$.*

Proof Since $0 \notin A_L^{\leq}$, then by Proposition 3.8 of (Boyadzhiev et al. 2018), $\{1_L\}$ is a minimal cover of L . Thus, every cover of L contains the element 1_L and so the cover $\{1_L\}$ is a refinement of every cover of L with

$$\text{ind}(\uparrow(1_L^* \vee 1_L)) = \text{ind}(\uparrow(0_L \vee 1_L)) = \text{ind}(\uparrow 1_L) = \text{ind}(\{1_L\}) = -1.$$

Therefore, $\text{ind}(L) = 0$. □

Proposition 11 *Let $(L, \leq)$ be a finite lattice, V be a minimal cover of L and $v \in V$. Then $\text{ind}(\uparrow(v \vee v^*)) = m$, where $m \in \{0, 1, \dots\}$, if and only if we apply Theorem 8 up to reach the lattice $\{x_n\}$ $m + 1$ times.*

Proof Let $m \in \{0, 1, \dots\}$. To compute the small inductive dimension $\text{ind}(\uparrow(v \vee v^*))$ we work in the following manner:

First Time: $\text{ind}(\uparrow(v \vee v^*)) \leq m \Leftrightarrow$ for every minimal cover V_1 of $\uparrow(v \vee v^*)$, $\text{ind}(\uparrow(v_1^* \vee v_1)) \leq m - 1$, for every $v_1 \in V_1$.

Second Time: For every $v_1 \in V_1$, $\text{ind}(\uparrow(v_1^* \vee v_1)) \leq m - 1 \Leftrightarrow$ for every minimal cover V_2 of $\uparrow(v_1 \vee v_1^*)$, we have $\text{ind}(\uparrow(v_2^* \vee v_2)) \leq m - 2$, for every $v_2 \in V_2$.

...

m-Time: For every $v_{m-1} \in V_{m-1}$, $\text{ind}(\uparrow(v_{m-1}^* \vee v_{m-1})) \leq 1 \Leftrightarrow$ for every minimal cover V_m of $\uparrow(v_{m-1} \vee v_{m-1}^*)$, we have $\text{ind}(\uparrow(v_m^* \vee v_m)) \leq 0$, for every $v_m \in V_m$.

m+1-Time: For every $v_m \in V_m$, $\text{ind}(\uparrow(v_m^* \vee v_m)) \leq 0 \Leftrightarrow$ for every minimal cover V_{m+1} of $\uparrow(v_m \vee v_m^*)$, we have $\text{ind}(\uparrow(v_{m+1} \vee v_{m+1}^*)) = -1$, for every $v_{m+1} \in V_{m+1}$. In this step, we state that $\text{ind}(\uparrow(v_{m+1} \vee v_{m+1}^*)) = -1$ if and only if $\uparrow(v_{m+1} \vee v_{m+1}^*) = \{x_n\}$, for every $v_{m+1} \in V_{m+1}$.

Therefore, $\text{ind}(\uparrow(v \vee v^*)) \leq m$ if and only if we apply Theorem 8 up to reach the lattice $\{x_n\}$ $m + 1$ times. From Step 1, that we get the result $m - 1$ for the corresponding small inductive dimensions, up to Step m , that we get the result 0 for the corresponding small inductive dimensions, we apply Theorem 8 m times. Finally, we apply it one more time to succeed to have the lattice $\{x_n\}$.

Moreover, since $\text{ind}(\uparrow(v \vee v^*)) \not\leq m - 1$, we have that $\text{ind}(\uparrow(v \vee v^*)) = m$ if and only if we apply Theorem 8 up to reach the lattice $\{x_n\}$ $m + 1$ times. □

Let $(L, \leq)$ be a finite lattice, V be a minimal cover of L and $v \in V$. In what follows, the procedure, which is described in the proof of Proposition 11 in order to compute the small inductive dimension $\text{ind}(\uparrow(v \vee v^*))$, will be called *minimal cover procedure*.

In Boyadzhiev et al. (2018), an algorithm which finds the set $MCov(L)$ of all minimal covers of an arbitrary finite lattice L is presented. Here, we combine this algorithm with Algorithms 1 and 2, Theorem 8, Propositions 9, 10 and 11 in order to succeed a matrix computation of the small inductive dimension of a finite lattice L .

In this algorithm, the individual part of Algorithms 2 and 3 seems to be the most suitable choice for our study. To compute the small inductive dimension it suffices to apply these algorithms only for the elements of the corresponding minimal covers.

Especially, if $(L, \leq)$ is a finite lattice, where $L = \{x_1, \dots, x_n\}$, with $n \geq 2$, then this procedure is presented as follows, named as Algorithm 3.

1. Find the order matrix $\mathbf{A}_L^{\leq}$ and the incidence matrix $T_L^{\leq}$ of L and go to Step 2.
2. If $0 \notin \mathbf{A}_L^{\leq}$, then print $\text{ind}(L) = 0$. Otherwise, go to Step 3.
3. Apply Algorithm 3.9 of (Boyadzhiev et al. 2018) to create the set $MCov(L)$.
 - If $MCov(L) = \{\{x_n\}\}$, then print $\text{ind}(L) = 0$.
 - Otherwise, put

$$MCov(L) = \{C_1^L, \dots, C_r^L\} \text{ and } C_i^L = \{x_{ji_1}^{i,L}, \dots, x_{ji_{f_i}}^{i,L}\},$$

for every $i \in \{1, \dots, r\}$ and go to Step 4.

4. Set $i = 1$ and go to Step 5.
5. Set $s = 1$ and go to Step 6.
6. Apply Algorithm 1 in order to find the element $(x_{jis}^{i,L})^*$ and go to Step 7.
7. Apply Algorithm 2 in order to find the lattice

$$\uparrow(x_{jis}^{i,L} \vee (x_{jis}^{i,L})^*) = \uparrow x_l.$$

If $\uparrow x_l = \{x_n\}$, then print $d_{i,s} = \text{ind}(\uparrow x_l) = -1$ and go to Step 9. Otherwise, go to Step 8.

8. Apply the minimal cover procedure for the lattice $\uparrow(x_{jis}^{i,L} \vee (x_{jis}^{i,L})^*)$ up to reach the lattice $\{x_n\}$. Print

$$d_{i,s} = \text{ind}(\uparrow(x_{jis}^{i,L} \vee (x_{jis}^{i,L})^*)) = w - 1,$$

where w is the number of the iterations that are needed in order to succeed to have the lattice $\{x_n\}$, and go to Step 9.

9. If $s < f_i$, then put $s \rightarrow s + 1$ (that is, replace the old s with $s + 1$) and go to Step 6. Otherwise, go to Step 10.
10. If $i < r$, then put $i \rightarrow i + 1$ (that is, replace the old i with $i + 1$) and go to Step 5. Otherwise, go to Step 11.
11. Print

$$\text{ind}(L) = \max\{d_{i,s} + 1 : i = 1, \dots, r, s = 1, \dots, f_i\}.$$

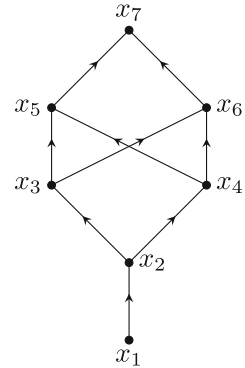
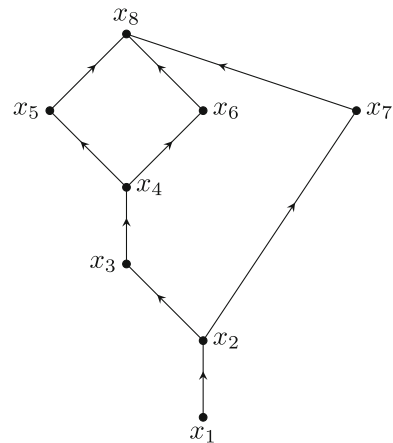
Example 4 (1) We consider the lattice $(L, \leq)$ which is represented in Fig. 3.

The incidence and order matrix of L are respectively

$$T_L^{\leq} = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}, \mathbf{A}_L^{\leq} = \begin{pmatrix} 1 & 2 & 2 & 2 & 2 & 2 & 2 \\ -2 & 1 & 2 & 2 & 2 & 2 & 2 \\ -2 & -2 & 1 & 0 & 2 & 2 & 2 \\ -2 & -2 & 0 & 1 & 2 & 2 & 2 \\ -2 & -2 & -2 & -2 & 1 & 0 & 2 \\ -2 & -2 & -2 & -2 & 0 & 1 & 2 \\ -2 & -2 & -2 & -2 & -2 & -2 & 1 \end{pmatrix}.$$

Following Algorithm 3 we have that

$$MCov(L) = \{C_1^L = \{x_3, x_4\}, C_2^L = \{x_5, x_6\}\}.$$

Fig. 3 The finite lattice $(L, \leq)$ **Fig. 4** The finite lattice $(M, \leq^*)$ 

We study the minimal cover $\mathcal{C}_1^L = \{x_3, x_4\}$. The set $\mathcal{C}_2^L$ is studied with the same way.

Following Algorithm 1 we have that $x_3^* = x_1$ and $x_4^* = x_1$. Moreover, following Algorithm 2 we have that

$$\uparrow(x_3 \vee x_3^*) = \{x_3, x_5, x_6, x_7\} \text{ and } \uparrow(x_4 \vee x_4^*) = \{x_4, x_5, x_6, x_7\}.$$

Also, for the lattice $\uparrow(x_3 \vee x_3^*)$ we apply only one time the minimal cover procedure and therefore, $d_{1,1} = 0$. Similarly, for the lattice $\uparrow(x_4 \vee x_4^*)$ we apply only one time the minimal cover procedure up and therefore, $d_{1,2} = 0$. Similarly, for the set $\mathcal{C}_2^L$, we get $d_{2,1} = d_{2,2} = 0$. Therefore,

$$\text{ind}(L) = \max\{d_{i,s} + 1 : i = 1, 2, 3 \text{ and } s = 1, 2\} = 1.$$

(2) We consider the lattice $(M, \leq^*)$ which is represented in Fig. 4.

The incidence and order matrix of M are respectively

$$T_M^{\leq*} = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

and

$$\mathbf{A}_M^{\leq*} = \begin{pmatrix} 1 & 2 & 2 & 2 & 2 & 2 & 2 & 2 \\ -2 & 1 & 2 & 2 & 2 & 2 & 2 & 2 \\ -2 & -2 & 1 & 2 & 2 & 2 & 0 & 2 \\ -2 & -2 & -2 & 1 & 2 & 2 & 0 & 2 \\ -2 & -2 & -2 & -2 & 1 & 0 & 0 & 2 \\ -2 & -2 & -2 & -2 & 0 & 1 & 0 & 2 \\ -2 & -2 & 0 & 0 & 0 & 0 & 1 & 2 \\ -2 & -2 & -2 & -2 & -2 & -2 & -2 & 1 \end{pmatrix}.$$

Following Algorithm 3 we have that

$$MCov(M) = \{C_1^M = \{x_3, x_7\}, C_2^M = \{x_5, x_6\}, C_3^M = \{x_5, x_7\}, C_4^M = \{x_6, x_7\}\}.$$

We study the minimal cover $C_1^M = \{x_3, x_7\}$. The sets C_2^M , C_3^M and C_4^M are studied with the same way.

Following Algorithm 1 we have that $x_3^* = x_1$ and $x_7^* = x_1$. Moreover, following Algorithm 2 we have that

$$\uparrow(x_3 \vee x_3^*) = \{x_3, x_4, x_5, x_6, x_8\} \text{ and } \uparrow(x_7 \vee x_7^*) = \{x_7, x_8\}.$$

Also, for the lattice $\uparrow(x_3 \vee x_3^*)$ we apply two times the minimal cover procedure up to the lattice $\{x_8\}$:

- for the first time we consider the minimal cover $\{x_5, x_6\}$ of the lattice $\uparrow(x_3 \vee x_3^*)$ and
- for the second time we consider the minimal cover $\{x_8\}$ of the lattices $\uparrow(x_5 \vee x_5^*)$ and $\uparrow(x_6 \vee x_6^*)$.

Therefore, $d_{1,1} = 1$.

For the lattice $\uparrow(x_7 \vee x_7^*)$ we apply only one time the minimal cover procedure up to the lattice $\{x_8\}$ (as the minimal cover of this lattice is $\{x_8\}$) and therefore, $d_{1,2} = 0$.

We continue with the same thinking, studying all the other minimal covers of M . We get $d_{2,1} = d_{2,2} = 0$, $d_{3,1} = d_{3,2} = 0$ and $d_{4,1} = d_{4,2} = 0$. Therefore,

$$\text{ind}(M) = \max\{d_{i,s} + 1 : i = 1, 2, 3, 4 \text{ and } s = 1, 2\} = 2.$$

5 Discussion

In this paper, we study the small inductive dimension of finite lattices using matrix algebra. Firstly, we investigate the notion of the pseudocomplement x^* of an element x of a finite lattice through incidence and order matrices and present an algorithm which finds this element. Then we study the lattice $\uparrow(x \vee x^*)$ using order matrices and present an algorithm which determines this lattice. Moreover, based on the notion of the minimal cover and the matrix characterizations of the pseudocomplement and the lattice $\uparrow(x \vee x^*)$, we provide an

algorithmic procedure which computes the small inductive dimension of an arbitrary finite lattice.

6 Conclusion

In this paper, we present an alternative approach to the small inductive dimension for the class of finite lattices using order and incidence matrices and algorithmic matrix procedures.

Acknowledgements The authors would like to thank the referees for the careful reading of the paper and the useful comments.

Data availability All data are mentioned in the paper.

Declarations

Conflict of interest Not applicable.

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